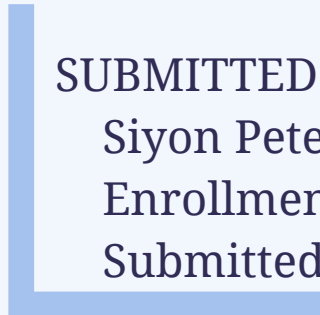




PROJECT REPORT

- PROJECT CODE: P1
- PROJECT TITLE: Drought Severity Assessment using Multi-temporal Vegetation Condition Index from Satellite Data.
- STUDY AREA: Kottayam, Kerala
- INSTITUTION: National Institute Of Technology, Calicut.



SUBMITTED BY,
Siyon Peter
Enrollment No:954558
Submitted on: 01/07/2026

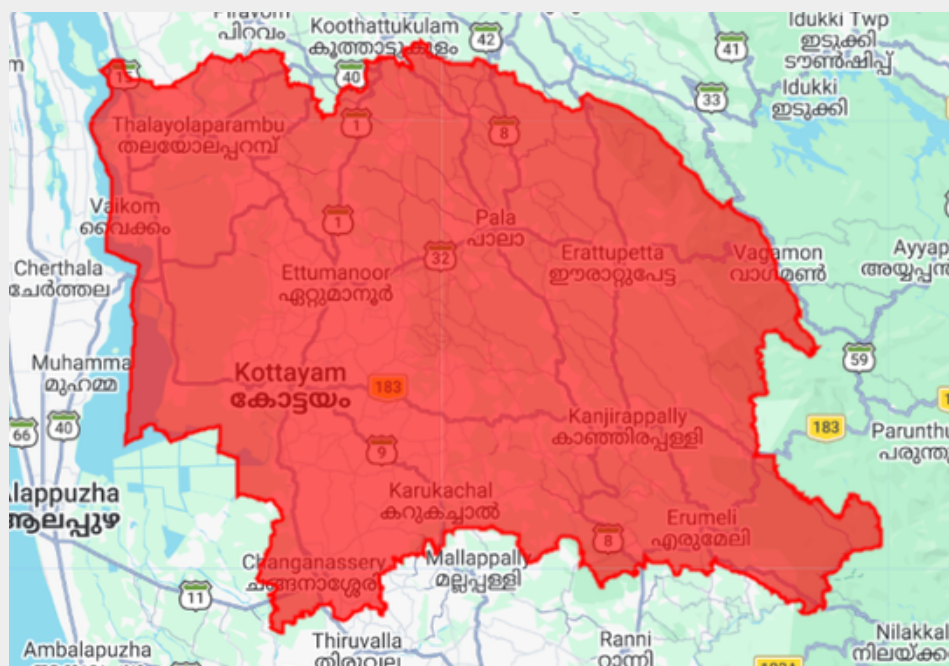


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Study Area	3
Data Used	4
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- **Title:** Drought Severity Assessment using Multi-temporal Vegetation Condition Index from Satellite Data.
- **Objective:** The objective of this project is to assess drought severity in Kottayam District, Kerala using multi-temporal MODIS NDVI satellite data and the Vegetation Condition Index (VCI). The study aims to identify vegetation stress zones and classify drought severity into different categories for effective environmental monitoring.
- **Study Area:** Kottayam District is situated in Kerala, precisely in the South-central part. The District is characterised by diverse land cover including agricultural fields, forests, wetlands, etc. Kottayam receives significant rainfall from both southwest and northeast monsoons, making vegetation monitoring important for agricultural planning.



Area of Interest(AOI): Kottayam, Kerala

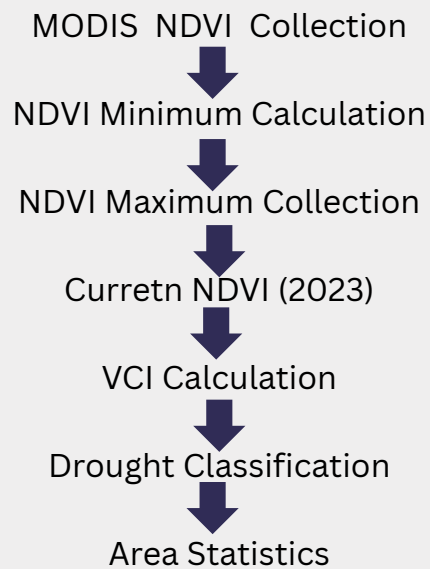


- **Data Used:**

Parameter	Details
Satellite Data	MODIS MOD13Q1
Parameter Used	NDVI
Time Period	2015 - 2023
Spatial Resolution	250m
Platform	Google Earth Engine
Boundary Data	Kerala District Boundary Shapefile

- **Methodology:**

The Workflow adopted for the assessment is as followa:



The Vegetation Condition Index (VCI) was calculated using:

$$VCI = \frac{NDVI - NDVI(min)}{NDVI(max) - NDVI(min)}$$

Drought Class Used:

VCI Range	Category
< 0.2	Extreme Drought
0.2 - 0.4	Moderate Drought
0.4 - 0.6	Mild Drought
> 0.6	Normal Vegetation

- **Result:**

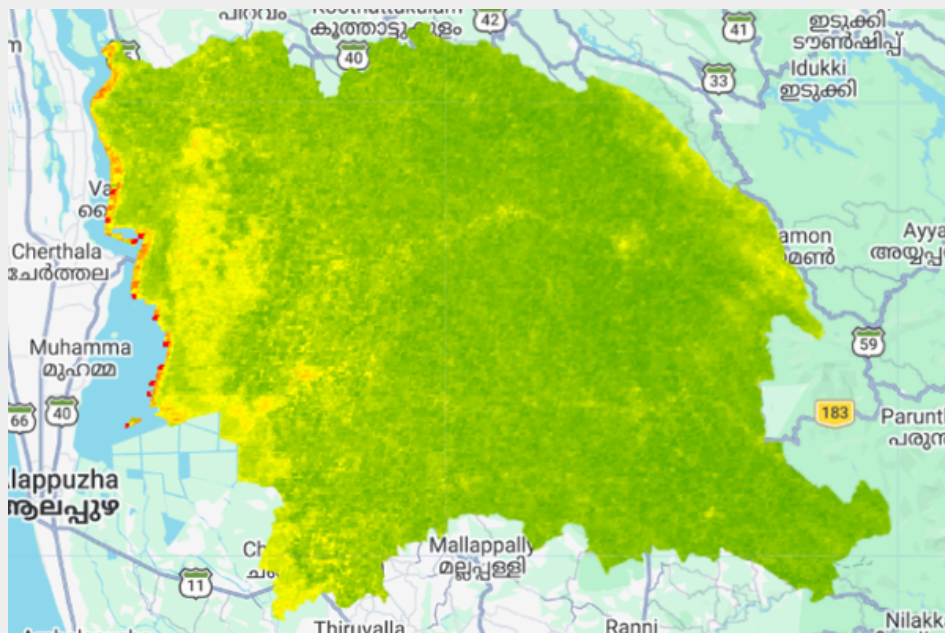
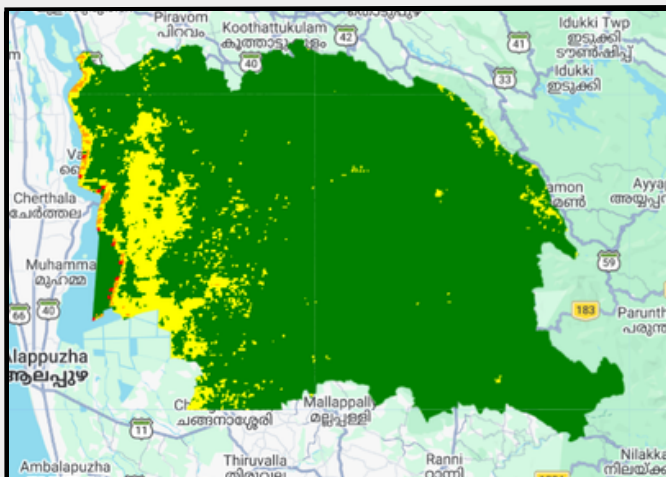


Figure 2. Vegetation Condition Index (VCI) map of Kottayam District





VCI Drought Classes

- VCI < 0.2 : Extreme Drought
- 0.2 - 0.4 : Moderate Drought
- 0.4 - 0.6 : Mild Drought
- VCI > 0.6 : Normal

Figure 3. Drought Severity Classification Based on VCI Values

```
Area Statistics (sq km)                                JSON
▼ Object (1 property)                                JSON
  ▼ groups: List (4 elements)
    ▼ 0: Object (2 properties)
      class: 1
      sum: 1.977270466161152
    ▼ 1: Object (2 properties)
      class: 2
      sum: 9.138124502251843
    ▼ 2: Object (2 properties)
      class: 3
      sum: 224.95037949413214
    ▼ 3: Object (2 properties)
      class: 4
      sum: 1965.6197972817743
```

Area Statistics of Drought Class

Class	Category	Area(sq.km)
1	Extreme Drought	1.97727
2	Moderate Drought	9.1381
3	Mild Drought	224.95037
4	Normal Vegetation	1965.6197



Interpretation:

The analysis indicates that most of the Kottayam District was covered by healthy vegetation during the study period. Approximately 1965.92 sq km was classified as Normal Vegetation, while only 1.98 sq km experienced extreme drought conditions. Mild drought conditions were experienced in scattered places.

• Conclusion

The study successfully utilized MODIS NDVI data and Google Earth Engine to assess drought severity in Kottayam District using the Vegetation Condition Index (VCI). The results show that the district largely maintained healthy vegetation conditions during 2023, with only limited areas experiencing drought stress. The methodology proved effective for large-scale drought monitoring and can support agricultural planning and environmental management activities. Limitations include the moderate spatial resolution of MODIS imagery and the influence of seasonal variations on vegetation conditions. Future studies may incorporate higher-resolution satellite datasets and rainfall information for improved accuracy.

• Reference:

1. MODIS MOD13Q1 VCI
2. Google Earth Engine Documentation

• APPENDIX A

Google Earth Engine Code

```
1
2 var kerala = ee.FeatureCollection({
3   "projects/project-1-497107/assets/Kerala_District_Boundary"
4 });
5
6
7 print('Properties', kerala.first());
8
9 var aoi = kerala.filter(
10   ee.Filter.eq('DISTRICT', 'Kottayam')
11 );
12
13
14 Map.centerObject(aoi, 9);
15 Map.addLayer(aoi, {color:'red'}, 'Kottayam');
16
17 var ndvi = ee.ImageCollection("MODIS/061/MOD13Q1")
18   .filterBounds(aoi)
19   .filterDate('2015-01-01', '2023-12-31')
20   .select('NDVI');
21
22 ndvi = ndvi.map(function(img){
23   return img.multiply(0.0001)
24     .copyProperties(img, ['system:time_start']);
25 });
26
27 var ndvi_min = ndvi.min();
28 var ndvi_max = ndvi.max();
29
30 var ndvi_current = ndvi
31   .filterDate('2015-01-01', '2023-12-31')
32   .mean();
33
34 var vci = ndvi_current
35   .subtract(ndvi_min)
36   .divide(ndvi_max.subtract(ndvi_min))
37   .rename('VCI');
38
```



```

39 Map.addLayer(
40   vci.clip(aoi),
41   {
42     min:0,
43     max:1,
44     palette:['red','yellow','green']
45   },
46   'VCI');
47
48 var drought = vci.expression(
49   "(b('VCI') < 0.2) ? 1" +
50   ": (b('VCI') < 0.4) ? 2" +
51   ": (b('VCI') < 0.6) ? 3" +
52   ": 4"
53 ).rename('Drought');
54
55 Map.addLayer(
56   drought.clip(aoi),
57   {
58     min:1,
59     max:4,
60     palette:['red','orange','yellow','green']
61   },
62   'Drought Class');
63
64 var areaImage = ee.Image.pixelArea()
65   .divide(1000000)
66   .addBands(drought);
67
68 var stats = areaImage.reduceRegion({
69   reducer: ee.Reducer.sum().group({
70     groupField: 1,
71     groupName: 'class'
72   }),
73   geometry: aoi.geometry(),
74   scale:250,
75   maxPixels:1e13
76 });
77 print('Area Statistics (sq.km)', stats);
78

```

```

Project 1 *
80 * Export.image.toDrive({
81   image: drought,
82   description: 'VCI_Drought_Map',
83   region: aoi.geometry(),
84   scale: 250,
85   maxPixels: 1e13
86 });
87
88
89
90 var legend = ui.Panel({
91   style: {
92     position: 'bottom-left',
93     padding: '8px 15px'
94   }
95 });
96
97
98 legend.add(ui.Label({
99   value: 'VCI Drought Classes',
100   style: {fontWeight: 'bold', fontSize: '14px'}
101 }));
102
103
104 var colors = ['red', 'orange', 'yellow', 'green'];
105 var names = [
106   'VCI < 0.2 : Extreme Drought',
107   '0.2 - 0.4 : Moderate Drought',
108   '0.4 - 0.6 : Mild Drought',
109   'VCI > 0.6 : Normal'
110 ];
111
112 for (var i = 0; i < colors.length; i++) {
113   legend.add(
114     ui.Panel([
115       ui.Label('', {
116         backgroundColor: colors[i],
117         padding: '8px',
118         margin: '0 5px 4px 0'
119       }),
120       ui.Label(names[i])
121     ], ui.Panel.Layout.Flow('horizontal'))
122   );
123 }
124 Map.add(legend);

```

